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GUARD ASSEMBLY FOR A CONVEYOR SYSTEM

Abstract

Various embodiments are directed to a guard assembly for a conveyor system. The conveyor system comprises a plurality of rollers rotatably disposed between a pair of side frames of the conveyor system. The guard assembly comprises a top cover and a holder such that the top cover is disposed in a gap between two adjacent rollers of the plurality of rollers of the conveyor system; the holder is detachably coupled to the top cover; the holder is configured to secure the top cover between the two adjacent rollers of the conveyor system and the top cover defines a plurality of friction-reducing features proximal to at least one roller of the conveyor system.

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Background/Summary

FIELD OF THE INVENTION

[0001] Various embodiments described herein relate generally to a material handling system for

handling objects and, more particularly, to apparatuses for use with a conveyor system configured to facilitate transportation of objects along a conveyor surface.

BACKGROUND

[0002] Generally, in material handling environments like, but not limited to, distribution centers, warehouses, inventories, or shipping centers, a material handling system can convey, handle, sort, and organize various type of articles (for example objects, cartons, cases, containers, shipment boxes, totes, packages, and/or the like) using one or more conveyor systems. Through applied effort, ingenuity, and innovation, many of these identified problems have been solved by developing solutions that are included in embodiments of the present disclosure, many examples of which are described in detail herein.

SUMMARY

[0003] Various embodiments are directed to a guard assembly for a conveyor system. In various embodiments, the conveyor system includes a plurality of rollers rotatably disposed between a pair of side frames of the conveyor system. In various embodiments, the guard assembly includes a top cover and a holder such that the top cover is disposed in a gap between two adjacent rollers of the plurality of rollers of the conveyor system; the holder is detachably coupled to the top cover; the holder is configured to secure the top cover between the two adjacent rollers of the conveyor system and the top cover defines a plurality of friction-reducing features proximal to at least one roller of the conveyor system.

[0004] In various embodiments, the friction-reducing features includes a plurality of recessed serrations. In various embodiments, the friction-reducing features includes a plurality of tapered grooves. In various embodiments, the friction-reducing features are equidistantly spaced-apart from each other.

[0005] In various embodiments, the holder includes a base wall with a first side and a second side; a first arm extending from the first side of the base wall and a second arm extending from the second side of the base wall such that the base wall is configured to support at least a portion of the top cover. In various embodiments, the holder includes at least one support rib extending between the first arm and the second arm. In various embodiments, the base wall includes a plurality of protrusions configured to prevent rotation of a fastener disposed therebetween.

[0006] Various embodiments are directed to a top cover for a guard assembly for conveyor system. In various embodiments, the conveyor system includes a plurality of rollers rotatably disposed between a pair of side frames of the conveyor system. In various embodiments, the top cover includes a bottom wall, a first sidewall and a second sidewall. In various embodiments, the bottom wall extends along a portion of a distance between the side frames of the conveyor system such that the bottom wall is in a gap between adjacent rollers of the plurality of rollers of the conveyor system. In various embodiments, the first sidewall and the second sidewall extend from bottom wall. In various embodiments, the first sidewall defines a plurality of friction-reducing features proximal to an adjacent roller of the conveyor system. In various embodiments, the second sidewall defines a plurality of friction-reducing features proximal to an adjacent roller of the conveyor system. In various embodiments, the friction-reducing features includes a plurality of recessed serrations. In various embodiments, the friction-reducing features includes a plurality of tapered grooves. In various embodiments, the friction-reducing features are equidistantly spaced-apart from each other.

[0007] In various embodiments, the first sidewall includes a first inclined wall and a first flange wherein the first inclined wall extends from the bottom wall and the first flange extends substantially horizontally from the first inclined wall.

[0008] In various embodiments, the second sidewall includes a second inclined wall and a second flange wherein the second inclined wall extends from the bottom wall and the second flange extends substantially horizontally from the second inclined wall.

[0009] In various embodiments, the bottom wall defines a receptacle configured to receive a head

of a fastener therein. In various embodiments, the bottom wall defines a first hole configured to receive a shank of a fastener therein.

[0010] In various embodiments, the bottom wall defines at least one pre-cut groove extending across a width of the bottom wall. In various embodiments, the pre-cut groove is configured to shear off due to a bending load applied to the top cover.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] The description of the illustrative embodiments can be read in conjunction with the accompanying figures. It will be appreciated that for simplicity and clarity of illustration, elements illustrated in the figures have not necessarily been drawn to scale. For example, the dimensions of some of the elements are exaggerated relative to other elements. Same numbers are used throughout the figures to reference like features and components. Embodiments incorporating teachings of the present disclosure are shown and described with respect to the figures presented herein, in which:

[0012] FIG. 1 illustrates a top view of a conveyor system, in accordance with various embodiments of the present disclosure;

[0013] FIG. 2 illustrates a bottom perspective view of a portion of the conveyor system, in accordance with various embodiments of the present disclosure;

[0014] FIG. 3A illustrates a front view of a guard assembly and two adjacent rollers of the conveyor system, in accordance with various embodiments of the present disclosure;

[0015] FIG. 3B illustrates a sectional view of the conveyor system taken along line A-A' of FIG. 1, in accordance with various embodiments of the present disclosure;

[0016] FIG. 4A illustrates a top perspective view of a holder of the guard assembly, in accordance with various embodiments of the present disclosure;

[0017] FIG. 4B illustrates a bottom perspective view of the holder of FIG. 4A, in accordance with various embodiments of the present disclosure;

[0018] FIG. 5A illustrates a top perspective view of a top cover of the guard assembly, in accordance with various embodiments of the present disclosure;

[0019] FIG. 5B illustrates a bottom perspective view of the top cover of FIG. 5A, in accordance with various embodiments of the present disclosure;

[0020] FIG. 5C illustrates a front view of the top cover of FIG. 5A, in accordance with various embodiments of the present disclosure; and

[0021] FIG. 6 illustrates a method of assembling a guard assembly on a conveyor system, according to an aspect of the present disclosure.

DETAILED DESCRIPTION

[0022] Indeed, various embodiments may be embodied in many different forms and should not be construed as limited to the embodiments set forth herein. Other embodiments, and modifications and improvements of the described embodiments, will occur to those skilled in the art and all such other embodiments, modifications and improvements are within the scope of the present disclosure. Features from one embodiment or aspect may be combined with features from any other embodiment or aspect in any appropriate combination. For example, any individual or collective features of method aspects or embodiments may be applied to apparatus, product or component aspects or embodiments and vice versa. Like numbers refer to like elements throughout.

[0023] It should be understood at the outset that, although illustrative implementations of one or more aspects are illustrated below, the disclosed assemblies, systems, and methods may be implemented using any number of techniques, whether currently known or not yet in existence. The disclosure should in no way be limited to the illustrative implementations, drawings, and

techniques illustrated below, but may be modified within the scope of the appended claims along with their full scope of equivalents. While values for dimensions of various elements are disclosed, the drawings may not be to scale.

[0024] The words “example,” or “exemplary,” when used herein, are intended to mean “serving as an example, instance, or illustration.” Any implementation described herein as an “example” or “exemplary embodiment” is not necessarily preferred or advantageous over other implementations. The phrases “in an embodiment,” “in some embodiments,” “according to one embodiment,” “in various embodiments” and the like generally mean that a particular feature, structure, or characteristic following the phrase may be included in at least one embodiment of the present disclosure or may be included in more than one embodiment of the present disclosure (importantly, such phrases do not necessarily refer to the same embodiment).

[0025] As used herein, directional terms used to describe a component, configuration, action, position, direction, and/or the like (for example, “rearward,” “an upward vertical direction,” “laterally outward,” “bottom,” “top,” and the like) are meant to be interpreted relative to a hypothetical conveyor system provided on an at least substantially flat horizontal surface, but the terms are not to be interpreted as requiring the conveyor system to be in this orientation at any particular time.

[0026] If the specification states a component or feature “may,” “can,” “could,” “should,” “would,” “preferably,” “possibly,” “typically,” “optionally,” “for example,” “often,” or “might” (or other such language) be included or have a characteristic, that specific component or feature is not required to be included or to have the characteristic. Such a component or feature may be optionally included in some embodiments, or it may be excluded.

[0027] Conveyor systems may be used in, for example, industrial manufacturing and packaging applications to facilitate the transportation of objects to a desired downstream location within a factory or a warehouse. For example, conveyor systems can include a plurality of cylindrical rollers disposed parallel and adjacent to each other along a length of the conveyor system. The plurality of rollers is arranged relative to one another such that the respective rolling surfaces of the rollers collectively define a conveyor path along which an object may be, at least, continuously transported towards the downstream location. Each roller can rotate about an axis thereof to help transfer objects disposed thereon to an adjacent roller and, in doing so, together, the plurality of cylindrical rollers supports in transferring the objects along a conveyor travel path towards an end of the conveyor system. Further, the conveyor systems include conveyor frame(s) having various structural components, such as, for example, but not limited to, sidewalls, panels, and/or the like, that are assembled relative to one another to define extremes capable of supporting the plurality of rollers therebetween.

[0028] In some examples, a guard apparatus may be installed within the conveyor system, by being secured relative to the side frames of the conveyor system, to prevent unauthorized and/or unintended access to an internal portion of the conveyor system, via a gap between any two adjacent rollers of the conveyor system. The guard apparatus may also prevent accumulation of debris, or small articles (such as fastener or coins) which may affect operation of the conveyor system. Further, the guard apparatus may also prevent accidental finger injury of an operator handling the objects on the conveyor system.

[0029] Additionally, the guard apparatus is subjected to repeated loading cycles, leading to localized deformations. Such deformations may cause a loosening of the guard apparatus installed within the conveyor system, which may result in undesirable movement thereof relative to the conveyor frame, undesirable noise (for example, rattling), partial inoperability of the guard apparatus, and/or unintentional disassembly of the guard apparatus relative to the conveyor system. Often, frictional force is developed between the rotating rollers and the conventional guard apparatus, resulting in reduced life and reliability thereof, undesirable loosening of the guard apparatus, and/or unintentional breakage of the guard apparatus. The conventional guard apparatus

is known to include complicated profiles configured to address issues resulting from the development of the friction as mentioned herein. However, such complicated profiles of the guard apparatus increase design complexity besides an increase in cost of manufacturing.

[0030] Various embodiments of the present disclosure are directed to a guard assembly for a conveyor system. The conveyor system includes a plurality of rollers rotatably disposed between a pair of side frames of the conveyor system. In various embodiments, the guard assembly includes a top cover and a holder, such that the top cover is disposed in a gap between two adjacent rollers of the plurality of rollers of the conveyor system. The holder is detachably coupled to the top cover and configured to secure the top cover between the two adjacent rollers. The top cover defines a plurality of friction-reducing features located proximal to at least one roller of the conveyor system.

[0031] FIG. 1 illustrates a top view of a conveyor system **100**, in accordance with various embodiments of the present disclosure. The conveyor system **100** includes a plurality of rollers **102** (such as a first roller **102-1**, a second roller **102-2**, a third roller **102-3**, and a fourth roller **102-4**), hereinafter alternatively, individually, and collectively referred to as “the roller(s) **102**” or “adjacent rollers **102-1**, **102-2**”, disposed between a pair of side frames **104-1**, **104-2** (hereinafter alternatively and collectively referred to as “the side frames **104**”), such that a gap **106** exists between any two adjacent rollers. For example, the gap **106** exists between two adjacent rollers **102-1**, **102-2** or **102-3**, **102-4**. In various embodiments, the adjacent rollers are joined by one or more driving members **103-1** and **103-2** (collectively referred to as “the driving member **103**”), such as a belt or an O-ring, at one end, where the driving member **103** is configured to transmit a rotary motion from one roller to the other, without slippage.

[0032] The conveyor system **100** may include multiple such driving members **103** configured to connect each pair of adjacent rollers to transmit the motion throughout a length of the conveyor system **100** in order to facilitate transfer of objects. As seen in FIG. 1, a guard assembly **108** is positioned within the conveyor system **100** to provide coverage over at least a portion of the gap **106** between the two adjacent rollers **102-1**, **102-2**. In some embodiments, the guard assembly **108** provides coverage over an entire portion of the gap **106** between the side frames **104-1** and **104-2**.

[0033] Referring to FIG. 2, a bottom perspective view of the conveyor system **100** is illustrated. FIG. 2 is described in conjunction with FIG. 1. The guard assembly **108** includes a top cover **202** and a holder **204**. The top cover **202** is disposed in the gap **106** between the two adjacent rollers **102-1**, **102-2** of the plurality of rollers **102**. The holder **204** is detachably coupled to the top cover **202** and configured to secure the top cover **202** between the two adjacent rollers **102-1**, **102-2**. In various embodiments, the top cover **202** extends between the side frames **104-1**, **104-2** and is configured to provide coverage over the entire portion of the gap **106** between the side frames **104-1**, **104-2**.

[0034] Referring to FIG. 3A, a front view of the guard assembly **108** and two adjacent rollers **102-1**, **102-2** of the conveyor system **100** is illustrated. FIG. 3A is described in conjunction with FIG. 1 and FIG. 2. The holder **204** is configured to detachably couple to the top cover **202** so as to secure the top cover **202** between the two adjacent rollers **102-1**, **102-2**. In some embodiments, a first fastener **302**, such as a screw or a bolt, defining a shank **302-1** and a head **302-2** is used in combination with a second fastener **304**, such as a nut, to secure the top cover **202** and the holder **204**. In some embodiments, the top cover **202** is disposed below the driving member **103**. As such, a top surface of the top cover **202** is located at a distance from the driving member **103**.

[0035] Referring to FIG. 3B, a cross-sectional view (taken along line A-A' of FIG. 1) of the guard assembly **108** disposed between the two adjacent rollers **102-1**, **102-2** is illustrated. The top cover **202** defines a plurality of friction-reducing features **306** proximal to at least one roller (such as the roller **102-1**) of the conveyor system **100**. In various embodiments, the friction-reducing features **306** include a plurality of recessed serrations. In various embodiments, the friction-reducing features **306** include a plurality of tapered grooves. In various embodiments, the friction-reducing features **306** are equidistantly spaced-apart from each other. In cases where the top cover **202** and

surface of the rollers **102-1**, **102-2** about during operation of the conveyor system **100**, the friction-reducing features **306** aid in reducing the contact surface area of the top cover **202** with the rollers **102-1**, **102-2**, thereby reducing friction therebetween.

[0036] Referring to FIG. 4A, a top perspective view of the holder **204** is illustrated. The holder **204** includes a base wall **402**, a first side **408**, a second side **410**, a first arm **404** extending from the first side **408** of the base wall **402**, and a second arm **406** extending from the second side **410** of the base wall **402**. The holder **204** further includes a support rib **412** extending between the first arm **404** and the second arm **406**. In an installed condition of the guard assembly **108**, the first arm **404** and the second arm **406** of the holder **204** are disposed proximal to the adjacent rollers **102-1** and **102-2**, respectively. As used herein, the term “installed condition” refers to the conveyor system **100** including the guard assembly **108** disposed in the gap **106** between the adjacent rollers **102-1** and **102-2**, as illustrated in FIG. 3A and FIG. 3B.

[0037] Referring to FIG. 4B, a bottom perspective view of the holder **204** is illustrated. FIG. 4B is described in conjunction with FIG. 3A and FIG. 4A. The base wall **402** includes a plurality of protrusions **414** configured to prevent rotation of the second fastener **304** disposed therebetween. For example, the plurality of protrusions **414** engages, through an interference fit, with side surfaces (not shown) of the second fastener **304**, such that the rotational motion of second fastener is restricted. Further, the base wall **402** defines a second hole **416** configured to receive the shank **302-2** of the first fastener **302** therethrough. In some embodiments, the first arm **404** of the holder **204** may include a first inclined portion **418** extending from the first side **408** of the base wall **402** and a second inclined portion **420** extending further divergently from the first inclined portion **418**. Similarly, the second arm **406** of the holder **204** may include a third inclined portion **422** extending from the second side **410** of the base wall **402** and a fourth inclined portion **424** extending further divergently from the third inclined portion **422**. In various embodiments, the second inclined portion **420** and the fourth inclined portion **424** are disposed proximal to the adjacent rollers **102-1** and **102-2**, respectively.

[0038] In various embodiments, the holder **204** may be made of light-weight polymer, such as plastics, manufactured through, for example, injection molding. To this end, the support rib **412** and the plurality protrusions **414** may be integrally formed together with the base wall **402**, the first arm **404** and the second arm **406**. In various embodiments, the holder **204** defines a hollow space “H” surrounded by the first arm **404**, the second arm **406** and one or more support ribs **412** joining the first arm **404** and the second arm **406**. Shape of the illustrated holder **204** should not be construed as limited. Instead, other shapes and features will be apparent to a person skilled in the art, albeit with few changes to the features described and illustrated herein. For example, in some embodiments, the holder **204** may have a bell-shape. Further, the holder **204** may include multiple fillets, chamfer, or edge radii.

[0039] FIG. 5A illustrates a top perspective view of the top cover **202** and FIG. 5B illustrates a bottom perspective view of the top cover **202**. FIG. 5A and FIG. 5B are described in conjunction with FIG. 3A. The top cover **202** includes a bottom wall **504**, a first sidewall **506** located on one side of the bottom wall **504** and a second sidewall **508** located on an opposite side of the bottom wall **504**. The bottom wall **504** extends along a portion of a distance between the side frames **104-1**, **104-2** of the conveyor system **100**. The bottom wall **504** is configured to be located in the gap **106** between adjacent rollers **102-1**, **102-2** of the plurality of rollers **102**. In various embodiments, the top cover **202** may be made up of light-weight polymer, such as plastics manufactured through, for example, injection molding. To this end, the bottom wall **504**, the first sidewall **506** and the second sidewall **508** are integrally molded to form the top cover **202**.

[0040] As shown in FIG. 5B, the top cover **202** defines the plurality of friction-reducing features **306** implemented as the plurality of recessed serrations. The bottom wall **504** may also define a receptacle configured to receive the head **302-1** of the first fastener **302** therein. The bottom wall **504** may define a first hole **516** configured to receive the shank **302-2** of the first fastener **302**

therein.

[0041] FIG. 5C illustrates a front view of the top cover **202** of FIG. 5A. According to an embodiment, the first sidewall **506** and the second sidewall **508** extends from the bottom wall **504**. The bottom wall **504** has a length “L” and a width “W”, where the length “L” corresponds to the portion of the distance between the side frames **104-1**, **104-2**, and the width “W” corresponds to the gap **106** between adjacent rollers **102-1**, **102-2** of the conveyor system **100**. The bottom wall **504** may define at least one pre-cut groove **518** extending across the width “W” thereof, where the pre-cut groove **518** is configured to shear off due to a bending load applied to the top cover **202**. In an event that a bending load is applied on the top cover **202**, the pre-cut groove exhibits a higher stress concentration as compared to rest of the top cover. As the load is increased, there is an eventual shearing off of the top cover **202** into multiple intact components with desired and reduced length. In such a manner, the length of the top cover **202** may be adjusted by shearing off the top cover **202** at pre-determined locations to fit into conveyor systems with, for example, varying width between the side frames **104-1**, **104-2**. In various embodiments, the base wall **402** of the holder **204** is configured to support at least a portion of the top cover **202**. For example, the bottom wall **504** of the top cover **202** abuts the base wall **402** of the holder **204** in the installed condition.

[0042] The first sidewall **506** of the top cover **202** includes a first inclined wall **524** extending from the bottom wall **504** and a first flange **526** extending substantially horizontally from the first inclined wall **524**. The second sidewall **508** of the top cover **202** further includes a second inclined wall **528** extending from the bottom wall **504** and a second flange **530** extending substantially horizontally from the second inclined wall **528**. The first flange **526** and the second flange **530** extend in opposite directions from the first inclined wall **524** and the second inclined wall **528** respectively. In various embodiments, the first inclined wall **524** and the second inclined wall **528** are disposed proximal to adjacent rollers **102-1**, **102-2** of the plurality of rollers **102**.

[0043] In some embodiments, the friction-reducing features **306** may be selectively provided on the first sidewall **506** and the second sidewall **508**. For example, the first sidewall **506** may define a plurality of friction-reducing features **306-1** and the second sidewall **508** may define a plurality of friction-reducing features **306-2**. In some embodiments, the friction-reducing features **306** may be defined on one of the first side wall **506** or the second sidewalls **508**. Specifically, the first flange **526** defines a plurality of friction-reducing features **306-3** proximal to an adjacent roller **102-1** of the conveyor system **100** and the second flange **530** defines a plurality of friction-reducing features **306-4** proximal to another adjacent roller **102-2** of the conveyor system **100**.

[0044] Shape of the illustrated top cover **202** should not be construed as limited. Instead, other shapes and features will be apparent to the person skilled in the art, albeit with few changes to the features described and illustrated herein. For example, the top cover **202** may include a U-shaped elongated wall with extending side flanges. Further, the top cover **202** may include multiple fillets, chamfer, or edge radii.

[0045] The present disclosure provides an improved guard assembly for a conveyor system. The guard assembly **108** as per the present disclosure prevents unauthorized and/or unintended access to portion of conveyor system **100**. This is achieved by, for example, the top cover **202** which provides coverage over the entire portion of the gap **106** between the side frames **104**. The guard assembly **108** also prevents accumulation of debris, or small article, such as fasteners or coins, which may lead to jamming of the conveyor system **100**. Further, the guard assembly **108** prevents finger injury by effectively, in a way, blocking the gap **106** between adjacent rollers **102-1**, **102-2**.

[0046] Various embodiments of the present disclosure provide further advantages. As the plurality of rollers **102** of the conveyor system **100** rotate in order to transport an object, the guard assembly **108** is configured to remain stationary. However, during the operation of the conveyor system **100**, the top cover **202** may abut surface of the rollers **102**. In such instances, the friction-reducing features **306** aid in reducing the contact surface area of the top cover **202** with the rollers **102-1**, **102-2**, thereby reducing friction therebetween.

[0047] Owing to the presence of the friction-reducing features **306**, the reliability and durable life of the guard assembly **108** may be increased. Further, in an event where a pull force is applied to remove or dismantle the guard assembly **108** from the conveyor system **100** in a vertical direction with respect to the rollers **102**, the first arm **404** and the second arm **406** of the holder **204** engage with the rollers **102** at their respective distal edges and prevent the guard assembly **108** from lifting or dislocation. Further, the protrusions **414** on the base wall **402** of the holder **204** prevent anti-rotation of the second fastener **304** disposed therein, thereby eliminating possibility of loosening out of the top cover **202** due to repeated loading cycles.

[0048] Advantageously, since the first inclined wall **524** of the first sidewall **506** and the second inclined wall **528** of the second sidewall **508** extend divergently from the bottom wall **504**, bending strength and bending stiffness of the top cover **202** may be improved, thereby reducing localized deformations. The support rib **412** adds to the strength and the stiffness of the holder **204**, thereby reducing localized deformations of the holder.

[0049] There are further technical advantages provided by the present disclosure. It may be desirable to manufacture the top cover **202** and the holder **204** using simpler methods, such as injection molding. The present disclosure provides a simplified structure for each of the top cover **202** and the holder **204** thereby reducing the manufacturing costs, etc.

[0050] There are still further advantages of the present disclosure. As per an embodiment, the length of the top cover **202** may be adjusted based on differing conveyor system dimensions owing to the pre-cut groove **518**.

[0051] Various aspects of the present disclosure provide a method of assembling the guard assembly **108** on the conveyor system **100**. FIG. **6** illustrates a method **600** of assembling the guard assembly **108** on the conveyor system **100**. FIG. **6** is described in conjunction with FIG. **1** through FIG. **5C**. The method includes placing the top cover **202**, from above the conveyor system **100**, such that the bottom wall **504** of the top cover **202** is disposed in the gap **106** between two adjacent rollers **102-1**, **102-2** of the plurality of rollers **102** of the conveyor system **100**; locating the holder **204** below the top cover **202**, from beneath the conveyor system **100**, such that the first hole **516** on the bottom wall **504** of the top cover **202** aligns with the second hole **416** on the base wall **402** of the holder **204**; inserting the first fastener **302**, such as, a screw or a bolt, through the first hole **516** of the top cover **202** and the second hole **416** of the holder **204**; tightening the first fastener **302** against the second fastener **304**, such as a nut, so that the top cover **202** and the holder **204** are securely held together between the pair of side frames **104**.

[0052] The foregoing method descriptions are provided merely as illustrative examples and are not intended to require or imply that the steps of the various embodiments must be performed in the order presented. As will be appreciated by one of skilled in the art, the order of steps in the foregoing embodiments may be performed in any order. Any reference to claim elements in the singular, for example, using the articles “a,” “an” or “the” is not to be construed as limiting the element to the singular.

[0053] When an element or layer is referred to as being “on,” “engaged to,” “connected to,” or “coupled to” another element or layer, it may be directly on, engaged, connected, or coupled to the other element or layer, or intervening elements or layers may be present. In contrast, when an element is referred to as being “directly on,” “directly engaged to,” “directly connected to,” or “directly coupled to” another element or layer, there may be no intervening elements or layers present. Other words used to describe the relationship between elements should be interpreted in a like fashion (for example, “between” versus “directly between,” “adjacent” versus “directly adjacent,” etc.). As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

[0054] Spatially relative terms, such as “inner,” “outer,” “beneath,” “below,” “lower,” “above,” “upper,” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. Spatially

relative terms may be intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures.

[0055] Many modifications and other embodiments of the present disclosure will be apparent to one skilled in the art to which the present disclosure pertain having the benefit of teachings presented in the foregoing descriptions and the associated drawings. Although the figures only show certain components of the system described herein, it is understood that various other components may be present. Therefore, it is to be understood that the present disclosure should not be limited to the specific embodiments disclosed and that modifications and other embodiments are intended to be included within the scope of the appended claims. Moreover, the steps in the method described above may not necessarily occur in the order depicted in the accompanying diagrams, and in some cases one or more of the steps depicted may occur substantially simultaneously, or additional steps may be involved. Although specific terms are employed herein, they are used in a generic and descriptive sense only and not for purposes of limitation.

Claims

1. A guard assembly for a conveyor system comprising a plurality of rollers rotatably disposed between a pair of side frames thereof, the guard assembly comprising: a top cover disposed in a gap between two adjacent rollers of the plurality of rollers; and a holder detachably coupled to the top cover, wherein the holder is configured to secure the top cover between the two adjacent rollers of the conveyor system; wherein the top cover defines a plurality of friction-reducing features proximal to at least one roller of the conveyor system.
2. The guard assembly of claim 1, wherein the friction-reducing features comprise a plurality of recessed serrations.
3. The guard assembly of claim 1, wherein the friction-reducing features comprise a plurality of tapered grooves.
4. The guard assembly of claim 1, wherein the friction-reducing features are equidistantly spaced apart from each other.
5. The guard assembly of claim 1, wherein the holder comprises: a base wall; a first arm extending from a first side of the base wall; and a second arm extending from a second side of the base wall, wherein the base wall is configured to support at least a portion of the top cover.
6. The guard assembly of claim 5, wherein the holder comprises at least one support rib extending between the first arm and the second arm.
7. The guard assembly of claim 5, wherein the base wall comprises a plurality of protrusions configured to prevent rotation of a fastener disposed therebetween.
8. A top cover for a guard assembly for a conveyor system comprising a plurality of rollers rotatably disposed between a pair of side frames thereof, the top cover comprising: a bottom wall extending along a portion of a distance between the side frames of the conveyor system, wherein the bottom wall is located in a gap between adjacent rollers of the plurality of rollers; a first sidewall extending from the bottom wall; and a second sidewall extending from the bottom wall; wherein at least one of the first sidewall or the second sidewall defines a plurality of friction-reducing features proximal to an adjacent roller of the conveyor system.
9. The top cover of claim 8, wherein the friction-reducing features comprise a plurality of recessed serrations.
10. The top cover of claim 9, wherein the recessed serrations comprise a plurality of tapered grooves.
11. The top cover of claim 8, wherein the friction-reducing features are equidistantly spaced apart from each other.
12. The top cover of claim 8, wherein the first sidewall further comprises: a first inclined wall extending from the bottom wall; and a first flange extending substantially horizontally from the

first inclined wall.

13. The top cover of claim 8, wherein the second sidewall further comprises: a second inclined wall extending from the bottom wall; and a second flange extending substantially horizontally from the second inclined wall.

14. The top cover of claim 8, wherein the bottom wall defines at least one receptacle configured to receive a head of a fastener therein.

15. The top cover of claim 8, wherein the bottom wall defines a first hole configured to receive a shank of a fastener therein.

16. The top cover of claim 8, wherein the bottom wall defines at least one pre-cut groove extending across a width of the bottom wall, the pre-cut groove configured to shear off due to a bending load applied to the top cover.
